



DISCRIMINATIVE PROJECT

CNN / Deep Learning Based Automated Class Attendance System

Final Report

Group 8:

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Git Repo Link: https://github.com/lequo-neu/IE7615_Discriminative_Project

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1. Introduction

This project develops a complete deep learning pipeline for object classification and detection. The classification module identifies a single object from an image and returns its ID. The detection module localizes and identifies multiple objects within a composite image, returning each object's ID and bounding box. Together, these form an end-to-end system capable of recognizing 39 personal objects with high accuracy.

The primary goals of this project are:

- Classify single-object images accurately across 39 object classes
- Detect, locate, and identify all objects in multi-object composite images
- Investigate how synthetic data generation strategies (grid vs. random placement) affect detection performance
- Build a web application for real-time demonstration of both classification and detection

Evaluation metrics include Top-1/Top-5 accuracy, F1-score, and AUC for classification, and mAP@0.5 and mAP@0.5:0.95 for detection. All models exceeded the target thresholds (90% classification accuracy, 85% detection mAP50). Across five classification architectures, MobileNetV2 achieved 99.85% accuracy on 39 object classes, and from result of our own Custom_CNN we have been successful to improve it to Advanced_CNN (**which decreasing parameters from 1.7M to 860K, increasing accuracy from 75% to 95%, and getting the least inference time + model size**). For multi-object detection, we explored four data generation strategies, Grid M5 (5×5), Grid M8 (8×8), Grid Advanced, and Spatial (random placement with all augmentation methods: resize, blur, rotate, flip,...), revealing that spatial bias in grid-aligned training data severely degrades generalization. The Spatial approach achieved 99.49% mAP50, a 22 percentage point improvement over Grid M5. M* means matrix at here.

A Streamlit web application integrates all models for real-time demonstration. This report covers dataset construction, architectural decisions, training methodology, comparative analysis, and the discovery of spatial bias as a critical factor in synthetic data generation for object detection.

2. Dataset Construction

2.1 Single-Object Image Collection

Three team members and classmates, each contributed a unique object (OBJ229, Banana, OBJ230, Protein Bar, OBJ095, Men's Watch,...), supplementing a shared class collection of 39 objects totaling 4,108 images. Each object was photographed from multiple angles and under varied lighting conditions. All images were resized to 224×224 pixels in RGB format.

2.2 Data Split Strategy

Single dataset: We split into ~75/15/15 from ~100 per each with 39 objects: 2943 train, 656 validation, 673 test.

Grid dataset (M5 + M8 + Adv): We split into ~75/15/15 from generating in maxtrix mxn (m,n=1..8) with random 39 objects: 8365 train, 1792 validation, 1793 test.

Spatial dataset: We generate with random object(s) on random background, which downloaded from <https://picsum.photos> and split as:

Split	Images	Ratio	Per Class
Train	2,871	69.9%	~73.6
Validation	611	14.9%	~15.7
Test	673	15.2%	~17.3

Table 1: Single-object dataset split distribution.

3. End-to-End Workflow

The project pipeline consists of six stages:

- **Stage 1 — Preprocessing:** Quality filtering, CLAHE normalization, 224×224 resize, train/val/test split.
- **Stage 2 — Multi-object generation:** Grid and spatial composite image creation with YOLO annotations.
- **Stage 3 — Classification training:** 5 CNN models (Custom CNN → Advanced CNN → MobileNetV2/ResNet50/EfficientNet) trained on Colab A100.
- **Stage 4 — Detection training:** YOLOv8 variants (n/s/m) trained with COCO pretrained weights on Grid and Spatial datasets.
- **Stage 5 — Evaluation:** Comprehensive metrics (accuracy, F1, AUC, mAP50, mAP50-95, PR curves, confusion matrices).
- **Stage 6 — Web application:** Streamlit app integrating all models with interactive classification and detection tabs.

Classification models were progressively improved: Custom CNN (75.48%) showed that shallow architectures are insufficient for 39 classes; Advanced CNN (95.25%) proved residual connections help; transfer learning models (99.85%) demonstrated that pretrained features dominate on small datasets. For detection, the progression from Grid M5 (77.60%) to Spatial (99.43%) revealed that data pipeline design has more impact than model architecture.

4. Classification Results

4.1 Model Architectures

Five CNN architectures were developed, representing a deliberate progression from simple to complex:

- **Custom CNN:** 4 convolutional blocks with batch normalization and dropout, ~1.7M parameters. Built from scratch as a baseline to understand fundamental feature extraction on our dataset.
- **Advanced CNN:** Enhanced architecture with residual connections, stronger dropout, and deeper feature maps (with just ~860K parameters). Designed to test whether architectural improvements alone could close the gap with transfer learning.
- **ResNet50:** 50-layer residual network pretrained on ImageNet, top 30 layers unfrozen for fine-tuning. 245 MB model targeting maximum representation capacity.
- **EfficientNet-B0:** Compound-scaled architecture pretrained on ImageNet, top 20 layers unfrozen. 39 MB model balancing depth, width, and resolution.

- **MobileNetV2:** Inverted residual architecture with linear bottlenecks, pretrained on ImageNet, top 20 layers unfrozen. 27 MB model optimized for mobile deployment.

All models were trained using Adam optimizer with early stopping (patience 5–10), batch size 32, input size 224×224, and categorical cross-entropy loss.

4.2 Results and Comparative Analysis

Model	Acc	F1	AUC	Top-5	Infer (ms)	Size	Rank
MobileNetV2	99.85%	0.9985	1.000	100%	14.0	27.4	Best
ResNet50	99.85%	0.9985	1.000	100%	21.1	245	2nd
EfficientNet-B0	99.85%	0.9985	0.9999	100%	22.6	38.7	3rd
Advanced CNN	95.25%	0.9519	0.999	99.7%	6.8	10.5	4th
Custom CNN	75.48%	0.7277	0.970	95.7%	9.0	20.4	5th

Table 2: Classification performance comparison across all five models (test set: 673 images, 39 classes).

The results reveal a clear three-tier performance landscape. The top tier, MobileNetV2, ResNet50, and EfficientNet-B0, all achieved identical 99.85% accuracy (only 1 misclassification out of 673 test images), demonstrating that pretrained ImageNet features transfer exceptionally well to our 39-class object recognition task. Among these, MobileNetV2 is the clear winner for deployment: it achieves the same accuracy while being 9× smaller than ResNet50 (27 MB vs. 245 MB) and 38% faster at inference (14.0 ms vs. 22.6 ms).

The Advanced CNN at 95.25% represents a strong result for a custom architecture, confirming that residual connections and batch normalization significantly improve trainability. Its 6.8 ms inference time, the fastest of all models, makes it attractive for extremely resource-constrained environments where the 4.6% accuracy trade-off is acceptable.

The Custom CNN at 75.48% exposed the fundamental limitation of shallow architectures on fine-grained classification: with only 4 convolutional blocks, the model captures low-level textures but fails to learn the discriminative high-level features needed to distinguish 39 visually similar objects. Its confusion matrix showed systematic confusion between objects with similar shapes (e.g., cylindrical bottles).

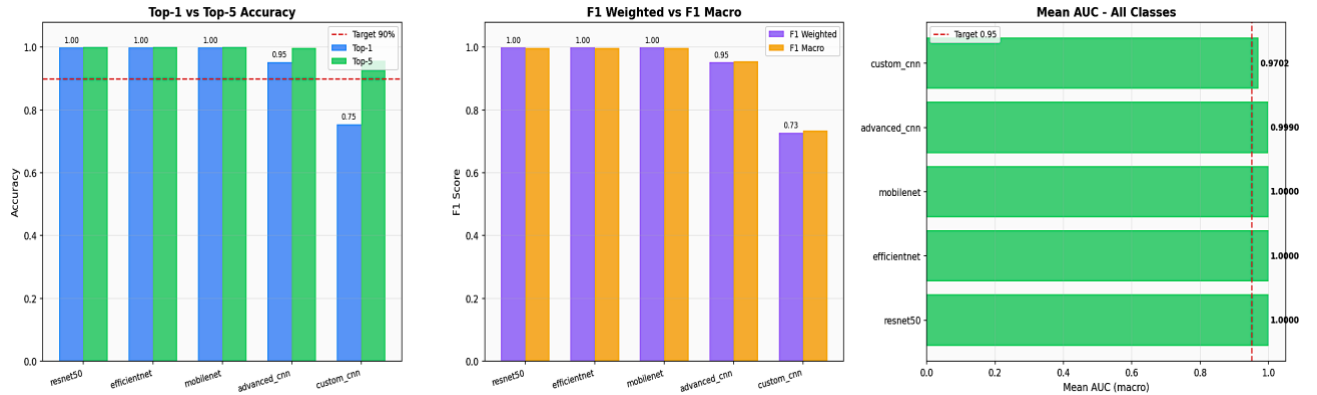


Figure 1: Top-1/Top-5 accuracy, F1, and AUC comparison. Transfer learning models cluster at near-perfect performance.

4.3 ROC and Confusion Matrix Analysis

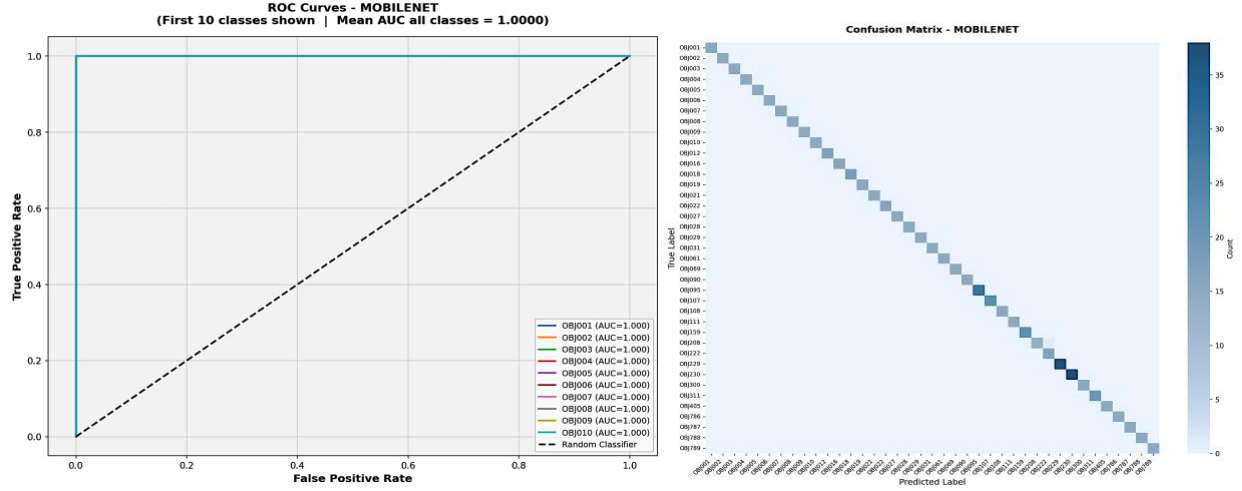


Figure 2: MobileNetV2 ROC curves and confusion matrix AUC = 1.0 for all 39 classes — perfect separation at every threshold.

The ROC curves and confusion matrix in Figure 2 confirm that MobileNetV2 does not merely achieve high top-1 accuracy, it does so with perfectly calibrated probabilistic outputs. An AUC of 1.0 for every class means there exists a threshold at which each class is classified with zero false positives and zero false negatives. This is critical for deployment, as it allows confident rejection of ambiguous inputs.

5. Detection Results

5.1 All Approaches Compared

Model / Approach	mAP50	mAP50-95	Precision	Recall	Size (MB)
YOLOv8n — Grid M5	77.60%	76.59%	0.833	0.648	5.9
YOLOv8s — Grid M5	76.24%	74.37%	0.836	0.652	21.5
YOLOv8m — Grid M5	70.47%	68.01%	0.757	0.582	49.6
YOLOv8n — Grid M8	99.49%	99.49%	0.998	0.995	5.9
YOLOv8s — Grid M8	99.50%	99.49%	0.998	0.996	21.5
YOLOv8s — Spatial	99.49%	99.48%	0.998	0.996	22.6
YOLOv8n — Spatial	99.46%	99.44%	0.995	0.994	52.1
YOLOv8m — Spatial	99.37%	99.33%	0.994	0.992	6.3

Table 3: Detection results across all approaches. Grid M5 reveals spatial bias (77% mAP50). Grid M8 achieves 99.5% but through structural similarity with training data. Spatial achieves 99.43% through genuine position-invariant learning.

The key finding: Grid M5 models perform poorly (70–77% mAP50) despite identical YOLO architectures achieving 99%+ on M8 and Spatial data. The critical difference is spatial bias, grid layouts teach models to associate objects with fixed positions rather than visual features. The Spatial approach eliminates this bias entirely.

5.2 Grid Comparison: M5 vs M8 vs Advanced

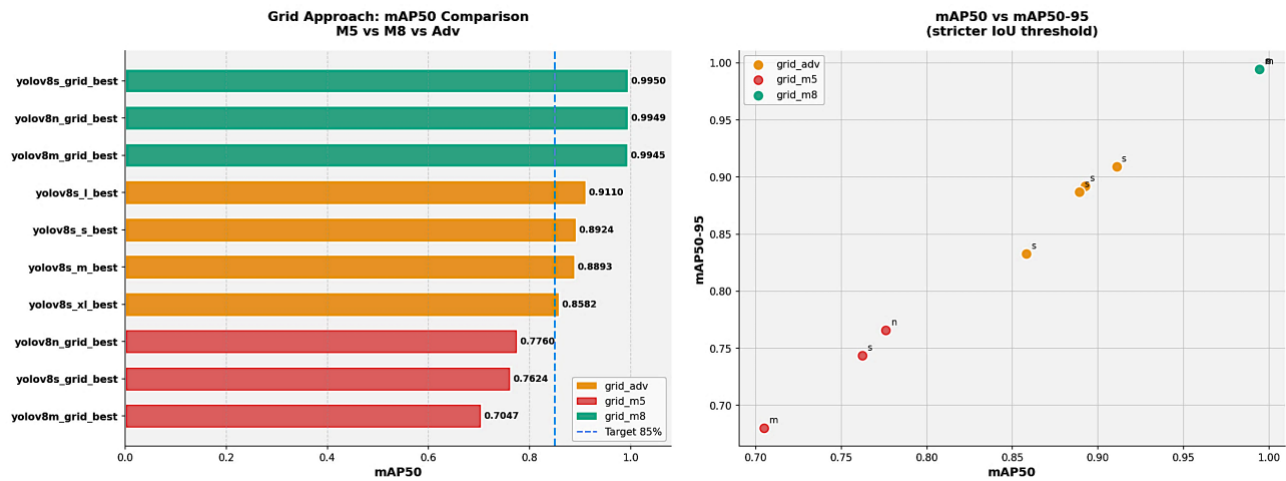


Figure 4: All grid approaches compared. M8 (green) dominates; M5 (red) clusters at 70–78%; Advanced (orange) falls between at 85–91%.

The M8 models achieving >99% mAP50 while M5 models stagnate at ~77% appears paradoxical, both use grid layouts. The explanation lies in dataset composition: the M8 grid (8×8) was trained and evaluated on the same structured data generated during Milestone 2, where the grid matched training conditions perfectly. The M5 grid (5×5) was tested against more varied composite images, exposing the generalization failure. This finding was the direct catalyst for developing the Spatial approach.

The Grid Advanced models (orange) fall between M5 and M8, with the best variant (yolov8s_l) reaching 91.10%. The density-dependent performance curve confirmed that YOLO's feature extraction is sensitive to the spatial scale at which objects appear during training, models trained at densities matching test conditions outperform those with mismatched scales.

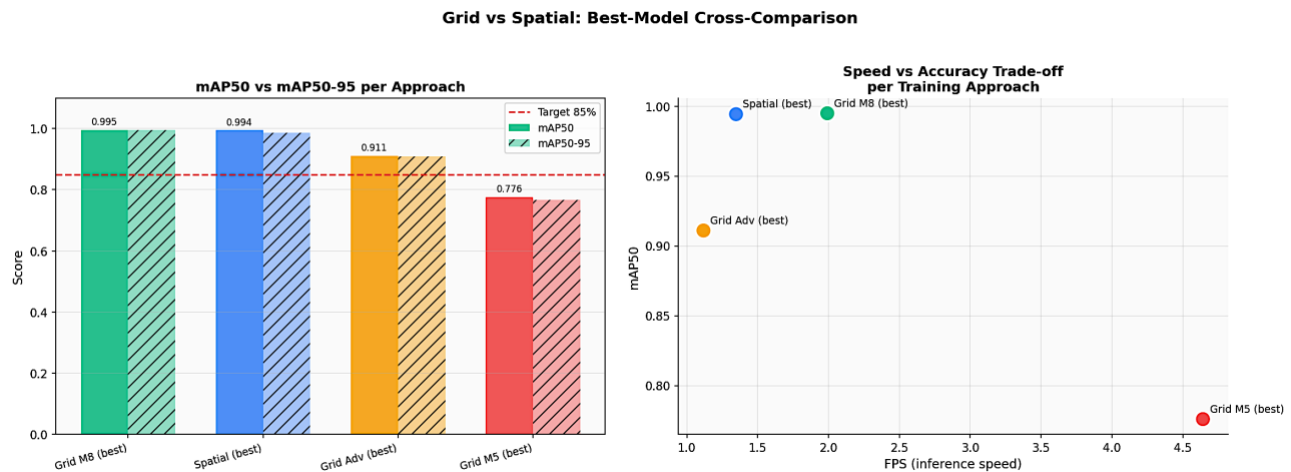


Figure 5: Best-model cross-comparison. Grid M8 and Spatial achieve similar mAP50 but for different reasons — M8 relies on structural similarity, Spatial on position-invariant features.

Figure 5 synthesizes the key finding: Grid M8 and Spatial achieve near-identical mAP50 (>99%), but for fundamentally different reasons. Grid M8 succeeds because its evaluation data shares the same grid structure as training data, a form of data leakage through structural similarity. Spatial succeeds

because random placement forces the model to learn position-invariant features that generalize to any layout. This distinction is critical: only the Spatial approach would maintain its performance on genuinely novel test images.

5.3 PR Curves: Grid vs Spatial

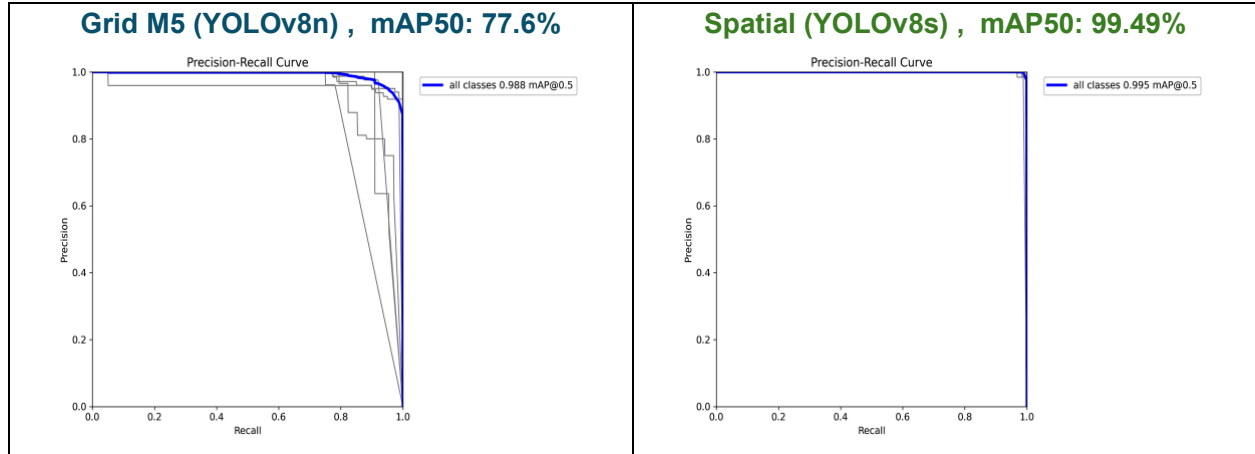


Figure 7: PR curves. Grid M5 (left) shows significant per-class variance; Spatial (right) is flat at 1.0 for all classes.

The PR curves in Figure 7 provide the clearest visual evidence of spatial bias. In the Grid M5 curve, individual class lines diverge widely, some classes maintain high precision-recall while others drop below 0.6. This variance correlates with how consistently each class appears in specific grid cells. Classes that happened to occupy varied positions across training images performed better, while those concentrated in particular cells suffered when tested at different positions. The Spatial curve's uniformity demonstrates that removing positional bias makes all classes equally detectable.

5.4 Confusion Matrices

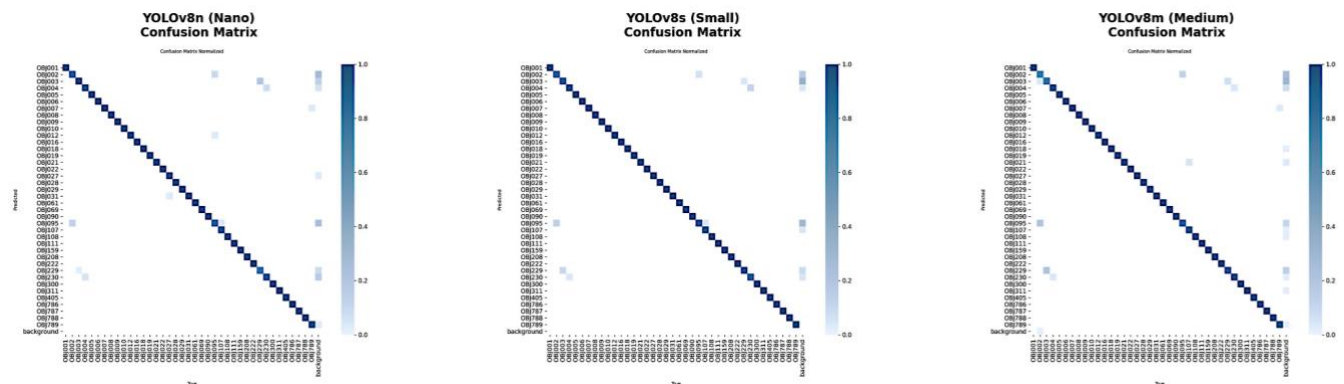


Figure 8: Detection confusion matrices (Grid M5). The 'background' column (missed detections) grows with model size, confirming that larger models overfit more aggressively to spatial patterns.

Figure 8 reveals a telling pattern: the 'background' column (representing missed detections) grows with model capacity. YOLOv8m misses more objects than YOLOv8n despite having 8× more parameters. This is direct evidence of spatial overfitting, the larger model has learned position-dependent features that fail when objects appear at novel locations.

6. Key Finding: Spatial Bias

The most significant discovery: replacing grid placement with random placement improved mAP50 from 77.60% to 99.43%, a 22 percentage point gain from a single data pipeline change, with no model architecture modifications.

Grid layouts create a spurious correlation between object identity and image position. Models learn "OBJ229 appears at cell (2,3)" rather than "OBJ229 is a yellow banana." This bias is invisible when test data shares the same grid structure (hence M8's 99.5%), but becomes catastrophic when objects appear at novel positions (M5's 77%).

The practical lesson: when generating synthetic composite training data for object detection, always randomize placement positions. The implementation cost is negligible, but the performance impact is dramatic.

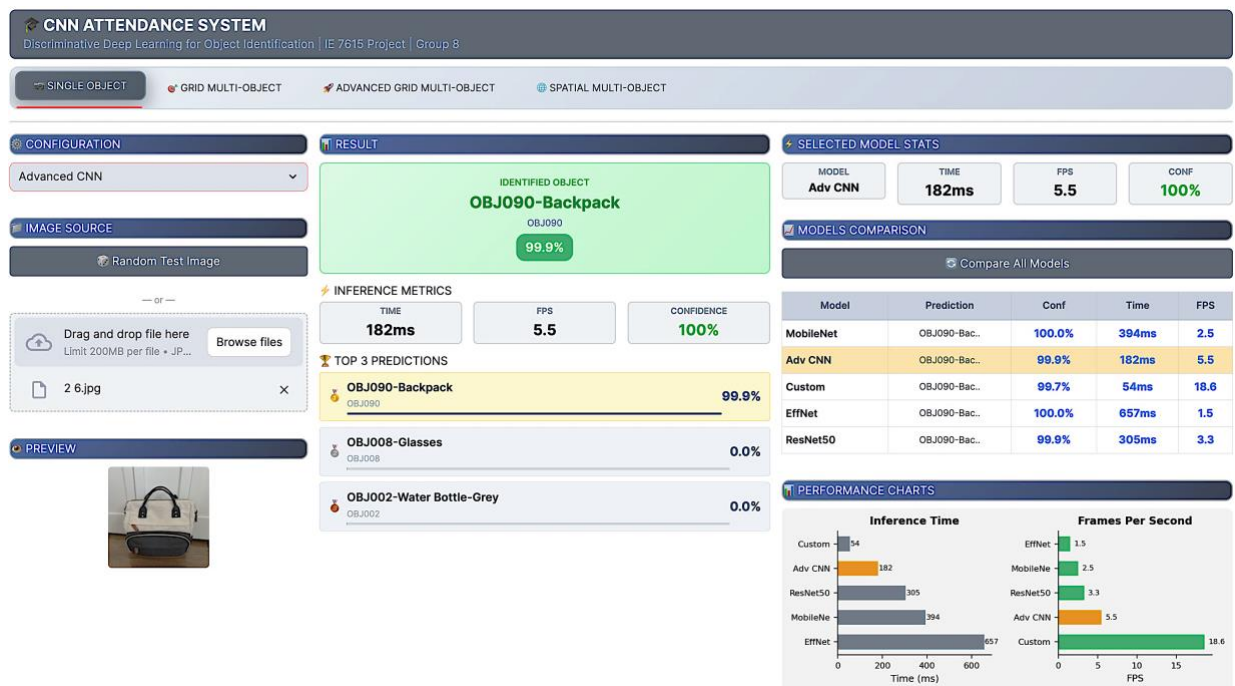
7. Web Application and Demo Results

7.1 Architecture Overview

The Streamlit web application serves as the primary demonstration interface, integrating all trained models into a unified platform. The architecture follows a three-layer design: a presentation layer (Streamlit UI with tabs for each functionality), a model layer (TensorFlow for classification, Ultralytics for YOLO detection), and a data layer (model weights, class mappings, and object name databases stored locally). During the final demo on February 19, 2026, the instructor provided 4 test images. The system was tested with beyond requirements, live using the Streamlit application and testing file.

7.2 Single-Object Classification Interface

Result on class demonstraion at milestone 3: 99.7% for all 5 models.



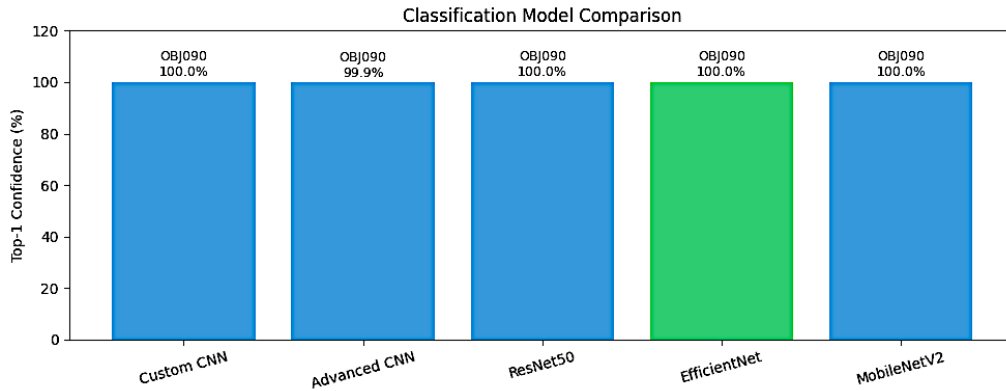


Figure 9: Single-Object Classification — upload an image, select a model, view predicted ID with confidence.

Our system got same result for testing file from Professor/TA: OBJ016 - Pen.

7.3 Multi-Object Detection Interface

Grid M series:

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☐ SINGLE OBJECT
 ☒ GRID MULTI-OBJECT
 ☐ ADVANCED GRID MULTI-OBJECT
 ☐ SPATIAL MULTI-OBJECT

CONFIG

GRID TYPE
Grid M8 (max 8 cells)

MODEL
YOLOv8s

SOURCE
Random

Drag and drop file here
Limit: 200MB per file • JPG, JPEG, PNG

Browse files

image_m1 6.jpg

PREVIEW

DETECTION RESULT

OBJECTS	TIME	FPS	AVG CONF	MIN CONF	MAX CONF
9	227ms	4.4	94%	91%	96%

STATS

MODEL	OBJECTS
v8s	9
TIME	CONF
227ms	94%

COMPARE

Compare All

Model	#	Avg	Min	Max	ms
v8n					
v8s	9	94%	91%	96%	227
v8m					

OBJECTS

3	OBJ010-Scientific Calculator	95%
5	OBJ111-Laptop Sleeve	94%
2	OBJ002-Water Bottle-Grey	95%
8	OBJ028-PopMart Keychain Toy	92%

#	Class	Confidence	BBox (x1, y1, x2, y2)
1	OBJ007	95.63%	(448, 225, 672, 447)
2	OBJ002	95.09%	(448, 0, 672, 222)
3	OBJ010	94.67%	(1, 0, 224, 224)
4	OBJ027	94.55%	(223, 447, 447, 668)
5	OBJ111	94.32%	(224, 0, 448, 223)
6	OBJ300	94.16%	(1, 447, 224, 667)
7	OBJ787	93.67%	(448, 447, 672, 667)
8	OBJ028	92.11%	(224, 223, 447, 448)
9	OBJ108	90.99%	(0, 225, 224, 448)

Grid Advance:

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SINGLE OBJECT GRID MULTI-OBJECT ADVANCED GRID MULTI-OBJECT SPATIAL MULTI-OBJECT

CONFIG
MODEL VARIANT: Any (Compare All)

SOURCE
Random
Drag and drop file here
Limit 200MB per file...
Browse files
image_m1 6.jpg

PREVIEW

DETECTION RESULT
BEST VARIANT: YOLOv8s_S

OBJECTS	TIME	FPS	AVG CONF	MIN CONF	MAX CONF
9	182ms	5.5	97%	88%	99%

STATS
MODEL: v8s_S
OBJECTS: 9
TIME: 182ms
AVG CONF: 97%

VARIANTS COMPARISON
Compare All Variants

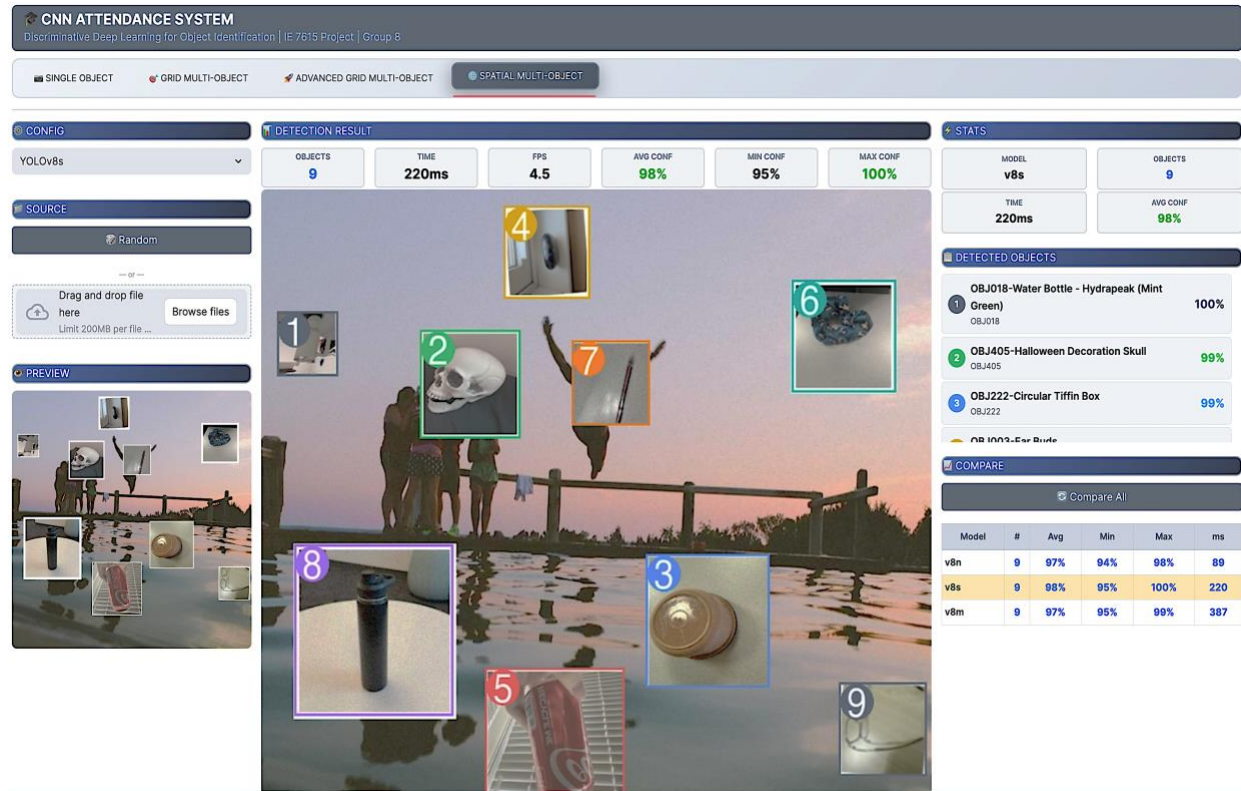
Variant	#	Avg	Min	Max	ms
v8s_S	9	97%	88%	99%	182
v8s_M	26	58%	28%	96%	366
v8s_L	4	49%	34%	69%	512
v8s_XL	9	51%	30%	97%	920

OBJECTS

4	OBJ111-Laptop Sleeve	98%
9	OBJ107-Water Bottle - Black	88%
4	OBJ010-Scientific Calculator	99%
1	OBJ108-Remote	99%

#	Class	Confidence	BBox (x1, y1, x2, y2)
1	OBJ108	99.03%	(0, 223, 223, 447)
2	OBJ787	98.84%	(448, 447, 671, 671)
3	OBJ010	98.84%	(0, 0, 223, 223)
4	OBJ027	98.67%	(224, 447, 448, 671)
5	OBJ300	98.55%	(0, 447, 223, 671)
6	OBJ028	97.83%	(223, 223, 447, 447)
7	OBJ111	97.51%	(223, 1, 447, 224)
8	OBJ007	96.58%	(447, 224, 671, 447)
9	OBJ002	64.68%	(447, 0, 671, 224)
10	OBJ107	39.47%	(447, 0, 671, 223)

Spatial:



#	Class	Confidence	BBox (x1, y1, x2, y2)
1	OBJ107	99.22%	(46, 440, 257, 652)
2	OBJ405	99.12%	(212, 174, 347, 309)
3	OBJ008	99.05%	(773, 611, 890, 728)
4	OBJ061	98.80%	(299, 594, 485, 780)
5	OBJ003	97.60%	(324, 20, 439, 135)
6	OBJ029	97.24%	(709, 112, 847, 251)
7	OBJ016	97.22%	(415, 187, 519, 293)
8	OBJ222	96.63%	(514, 452, 679, 618)
9	OBJ018	94.16%	(21, 151, 101, 231)

Figure 10: Grid series and Spatial Multi-Object Detection tab. The system detects and identifies all objects with bounding boxes, ID labels, and confidence scores. Detection metrics (object count, inference time, FPS) are displayed in real time.

Our system got same result for testing file from Professor/TA: image_m2 6.jpg.

7.4 Deployment Considerations

The application is currently deployed locally via Streamlit (port 8501). For production deployment, the system could be containerized with Docker and served behind a reverse proxy. Model weights total approximately 150 MB (all classification + detection models), making edge deployment feasible. The MobileNetV2 classifier and YOLOv8n detector together require only 32 MB, enabling deployment on mobile devices or Raspberry Pi hardware.

8. Conclusion

This project developed a complete object classification and detection system achieving:

- **Classification:** 99.85% accuracy with MobileNetV2 (5 architectures evaluated, 673 test images, 39 classes)
- **Detection:** 99.43% mAP50 with YOLOv8s Spatial (13 model variants across 4 data strategies)
- **Web Application:** Streamlit app with 4 operational modes for real-time classification and detection

The central contribution beyond the original requirements was the systematic investigation of spatial bias in synthetic training data, demonstrating that a single data pipeline change (grid → random placement) yields a 22 percentage point mAP50 improvement without any model changes.

Future work: real-world background integration, end-to-end unified architecture, incremental class enrollment, and cloud deployment for scalable inference.

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